

MATHEMATICS

SCIENTIFIC STREAM

Complete Upper Secondary / Terminale Coursebook

English explanations - worked examples - exercises - revision

$$f'(x) = \lim [f(x+h)-f(x)]/h$$

$$a_n = a_1 + (n-1)d$$

$$\sin^2 x + \cos^2 x = 1$$

$$P(A | B) = P(AB) / P(B)$$

80-PAGE SCIENTIFIC MATHEMATICS

Designed for systematic study, classroom use and examination revision.

How to Use This Book

COURSE ORIENTATION

LEARN

Read the concept explanation first. Focus on definitions, notation and the conditions under which a formula may be used.

WATCH THE METHOD

Worked examples show the reasoning order: identify the structure, choose a method, perform algebra carefully and check the answer.

PRACTISE

Complete the six practice questions at the end of every lesson. The first three have quick answers near the end of the book.

REVISE

Use the mixed reviews and mock examinations after finishing several chapters. Return to weak topics using the contents page.

Scientific problem-solving routine

1. Translate the question into mathematical quantities and notation.
2. Identify what is known, what is unknown and which conditions apply.
3. Choose a theorem, formula or method that matches the structure.
4. Work line by line and keep exact values as long as possible.
5. Check units, signs, domain restrictions and whether the result is reasonable.

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COURSE MAP

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REVISION & REFERENCE Mixed Reviews pp.66-70 | Mock Exams pp.71-74 | Answers pp.75-77 | Formula Sheet p.78

Notation, Symbols & Calculator Habits

COURSE ORIENTATION

Symbol	Meaning	Example
R	real numbers	$x \in R$ means x is real
	belongs to	$3 \in N$
	not equal	$x \neq 0$
	approximately equal	$\pi \approx 3.14159$
	infinity	$x \rightarrow \infty$
Σ	summation	Σx_i
Δ	change/discriminant	$\Delta = b^2 - 4ac$
	integral	$\int f(x) dx$
	square root	$\sqrt{49} = 7$
$^\circ$	degrees	$90^\circ = \pi/2$

CALCULATOR DISCIPLINE

Check degree/radian mode before trigonometry. Use brackets around complete numerators and denominators. Keep at least 4-6 significant digits during intermediate steps, then round only at the end.

EXACT VS APPROXIMATE

Prefer exact forms such as 2 , $\pi/3$ and $\ln 5$ when the question does not request decimals. Decimal approximations are useful for interpretation, measurement and numerical modelling.

Lesson 1: Algebraic Expressions & Factorisation

ALGEBRA FOUNDATIONS

LEARNING GOALS

- Understand the central ideas and notation of algebraic expressions & factorisation.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Algebraic manipulation is the language of advanced mathematics. You should be able to expand, collect like terms, factorise and simplify expressions before solving equations or analysing functions.

KEY FORMULAS

$a^2 - b^2 = (a-b)(a+b)$	difference of two squares
$(a+b)^2 = a^2+2ab+b^2$	square of a sum
$a^3-b^3=(a-b)(a^2+ab+b^2)$	difference of cubes

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 1: Worked Practice

ALGEBRA FOUNDATIONS

WORKED EXAMPLE

Take out the greatest common factor first, then check whether more factorisation is possible.

Factorise $6x^2 - 15x$

1. Common factor is $3x$

2. $6x^2 - 15x = 3x(2x - 5)$

Answer: $3x(2x - 5)$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Expand $(2x-3)(x+5)$.
2. Factorise x^2-9 .
3. Factorise $4x^2+12x+9$.
4. Simplify $(x^2-4)/(x-2)$, $x \neq 2$.
5. Factorise $2x^3-8x$.
6. Find the coefficient of x in $(x+2)(3x-1)$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 2: Linear & Quadratic Equations

EQUATIONS

LEARNING GOALS

- Understand the central ideas and notation of linear & quadratic equations.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

An equation states that two expressions are equal. Solving means finding every value that makes the statement true. Quadratic equations may be solved by factorisation, completing the square or the quadratic formula.

KEY FORMULAS

$ax+b=0 \Rightarrow x=-b/a$ linear equation, $a \neq 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ quadratic formula

$\Delta = b^2 - 4ac$ discriminant

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
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Lesson 2: Worked Practice

EQUATIONS

WORKED EXAMPLE

Use factorisation or the quadratic formula. Factorisation is quickest here.

Solve $2x^2 - 5x - 3 = 0$

1. $2x^2 - 6x + x - 3 = 0$

2. $(2x+1)(x-3)=0$

Answer: $x=3$ or $x=-1/2$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Solve $5x-17=3$.
2. Solve $x^2-7x+12=0$.
3. Solve $3x^2+2x-1=0$.
4. Find k if $x=2$ solves $x^2+kx-8=0$.
5. State number of real roots of $x^2+4x+8=0$.
6. Complete the square: x^2+6x+1 .

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 3: Systems of Equations & Matrices

LINEAR ALGEBRA

LEARNING GOALS

- Understand the central ideas and notation of systems of equations & matrices.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Simultaneous equations describe quantities that must satisfy several conditions at the same time. Matrices provide a compact way to organise coefficients and are useful for elimination, transformations and modelling.

KEY FORMULAS

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	2x2 matrix
$\det(A) = ad - bc$	determinant
$A^{-1} = (1/\det A) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$	inverse if $\det(A) \neq 0$

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 3: Worked Practice

LINEAR ALGEBRA

WORKED EXAMPLE

Add equations to eliminate y .

Solve $x+y=7$ and $2x-y=5$

1. $3x=12$ so $x=4$

2. Substitute into $x+y=7$: $y=3$

Answer: $(x,y)=(4,3)$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Solve $2x+y=9$ and $x-y=0$.
2. Find $\det\left(\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}\right)$.
3. Multiply $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ by $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$.
4. Find inverse of $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.
5. Solve $3x+2y=7$, $x-y=1$.
6. Explain when a 2×2 matrix has no inverse.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 4: Inequalities & Absolute Value

ALGEBRA FOUNDATIONS

LEARNING GOALS

- Understand the central ideas and notation of inequalities & absolute value.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Inequalities compare quantities rather than declaring exact equality. When multiplying or dividing by a negative number, reverse the inequality sign. Absolute value represents distance from zero.

KEY FORMULAS

$ x < a \Rightarrow -a < x < a$	inside an interval
$ x > a \Rightarrow x < -a \text{ or } x > a$	outside an interval
$(x-r)(x-s) > 0$	use sign analysis

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 4: Worked Practice

ALGEBRA FOUNDATIONS

WORKED EXAMPLE

Convert the absolute-value inequality into a compound inequality.

Solve $|2x-3| \leq 5$

1. $-5 \leq 2x-3 \leq 5$

2. $-2 \leq 2x \leq 8$, so $-1 \leq x \leq 4$

Answer: $[-1, 4]$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Solve $3x-2 > 10$.
2. Solve $-2x+5 \leq 1$.
3. Solve $|x-4| < 3$.

4. Solve $x^2-5x+6 > 0$.

5. Represent $x \leq 2$ or $x > 3$ on a number line.

6. Solve $|3x+1| \leq 7$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 5: Polynomial Functions

FUNCTIONS

LEARNING GOALS

- Understand the central ideas and notation of polynomial functions.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A polynomial function is built from non-negative integer powers of x . Its roots, factors, degree and leading coefficient help describe its graph and long-term behaviour.

KEY FORMULAS

$P(r)=0 \Leftrightarrow (x-r)$ is a factor

factor theorem

remainder= $P(a)$

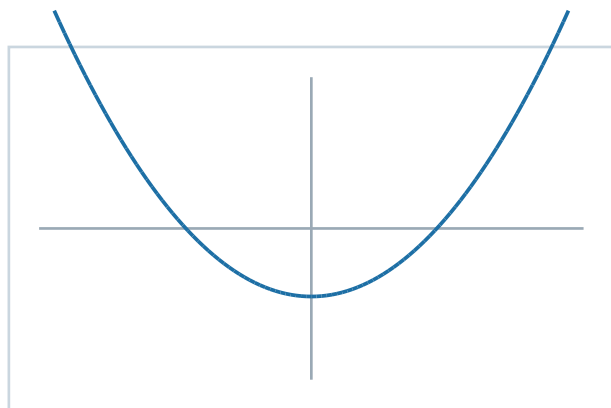
remainder theorem

degree $n \Rightarrow$ at most n real roots

root count

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.



GRAPH READING

Read intercepts, turning points, asymptotes, domain/range and end behaviour directly from the graph. Connect each visible feature with its algebraic cause.

Lesson 5: Worked Practice

FUNCTIONS

WORKED EXAMPLE

Group terms to reveal common factors.

Factor $P(x)=x^3-4x^2-x+4$

1. $x^2(x-4)-1(x-4)$

2. $(x^2-1)(x-4)=(x-1)(x+1)(x-4)$

Answer: roots: -1, 1, 4

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find $P(2)$ for $P=x^3-3x+1$.
2. Test whether $x-2$ divides x^3-5x+6 .
3. Factor $x^3-3x^2-4x+12$.
4. State end behaviour of $-2x^4+x$.
5. Find a cubic with roots 1,2,3.
6. Sketch $y=(x-1)(x+2)$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 6: Rational Functions

FUNCTIONS

LEARNING GOALS

- Understand the central ideas and notation of rational functions.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A rational function is a quotient of polynomials. Domain restrictions, intercepts and asymptotes are essential. Simplifying an expression does not remove an excluded value from the original domain.

KEY FORMULAS

$$f(x)=P(x)/Q(x), Q(x) \neq 0$$

domain restriction

$$Q(x)=0$$

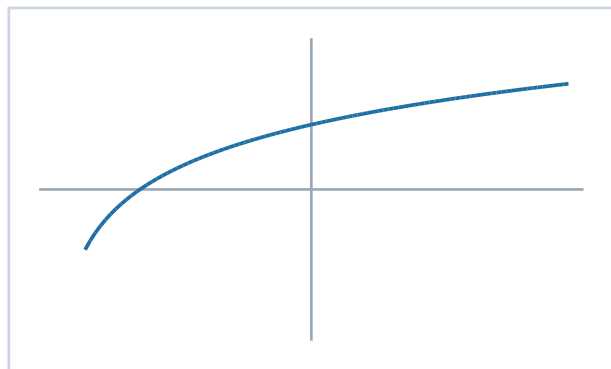
candidate vertical asymptote

$$\deg P < \deg Q \Rightarrow y=0$$

horizontal asymptote

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.



GRAPH READING

Read intercepts, turning points, asymptotes, domain/range and end behaviour directly from the graph. Connect each visible feature with its algebraic cause.

Lesson 6: Worked Practice

FUNCTIONS

WORKED EXAMPLE

Identify domain, intercepts and asymptotes.

Analyse $f(x)=(x+1)/(x-2)$

1. Domain: $x \neq 2$; vertical asymptote $x=2$
2. x-intercept -1; y-intercept $-1/2$

Answer: VA $x=2$, HA $y=1$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. State domain of $(x-3)/(x^2-9)$.
2. Find vertical asymptote of $2/(x+4)$.
3. Find x-intercept of $(x-5)/(x+1)$.
4. Simplify $(x^2-1)/(x-1)$.
5. Find horizontal asymptote of $(3x+1)/(x-2)$.
6. Sketch $1/x$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 7: Exponents & Radicals

ALGEBRA FOUNDATIONS

LEARNING GOALS

- Understand the central ideas and notation of exponents & radicals.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Exponent laws compress repeated multiplication and extend naturally to negative and fractional powers. Radical expressions can often be rewritten as rational exponents to simplify calculations.

KEY FORMULAS

$a^m a^n = a^{(m+n)}$	product law
$a^{-n} = 1/a^n$	negative exponent
$a^{(p/q)} = \text{qth_root}(a^p)$	fractional exponent

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 7: Worked Practice

ALGEBRA FOUNDATIONS

WORKED EXAMPLE

Use the denominator as the root and numerator as the power.

Simplify $27^{(2/3)}$

1. cube root of 27 is 3
2. $3^2=9$

Answer: 9

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Simplify $x^3 \cdot x^5$.
2. Rewrite $1/x^4$ using negative powers.
3. Evaluate $16^{(3/4)}$.
4. Simplify $\sqrt{50}$.
5. Rationalise $3/\sqrt{5}$.
6. Solve $2^{(x+1)}=16$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 8: Logarithms

FUNCTIONS

LEARNING GOALS

- Understand the central ideas and notation of logarithms.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A logarithm answers the question: to what exponent must the base be raised? Logarithms turn products into sums and powers into multipliers, making them essential in exponential equations and scientific models.

KEY FORMULAS

$\log_b(x)=y \Leftrightarrow b^y=x$	definition
$\log_b(xy)=\log_b x+\log_b y$	product rule
$\log_b(x^k)=k \log_b x$	power rule

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 8: Worked Practice

FUNCTIONS

WORKED EXAMPLE

Take logarithms of both sides.

Solve $3^x = 20$

1. $x \ln 3 = \ln 20$

2. $x = \ln 20 / \ln 3$

Answer: $x \approx 2.727$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Convert $\log_2 8 = 3$ to exponential form.
2. Simplify $\log 1000$.
3. Expand $\ln(3x^2)$.
4. Combine $\ln x + \ln(x-1)$.
5. Solve $5^x = 12$.
6. State domain of $\ln(x-4)$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 9: Functions & Transformations

FUNCTIONS

LEARNING GOALS

- Understand the central ideas and notation of functions & transformations.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A function assigns exactly one output to each allowed input. Transformations let you predict a new graph from a known one without plotting many points.

KEY FORMULAS

$$y=f(x)+k$$

vertical shift by k

$$y=f(x-h)$$

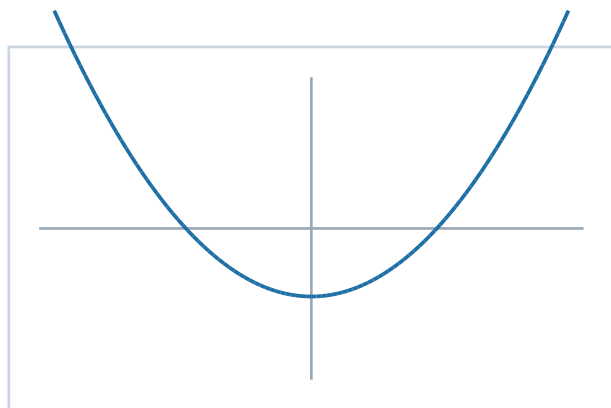
horizontal shift right h

$$y=af(x)$$

vertical stretch by $|a|$

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.



GRAPH READING

Read intercepts, turning points, asymptotes, domain/range and end behaviour directly from the graph. Connect each visible feature with its algebraic cause.

Lesson 9: Worked Practice

FUNCTIONS

WORKED EXAMPLE

Compare with $y=x^2$.

Describe $y=(x-2)^2+3$

1. $x-2$ shifts the parabola right by 2
2. $+3$ shifts it up by 3

Answer: vertex (2,3)

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. State domain of $\sqrt{x-1}$.
2. Describe $y=(x+4)^2$.
3. Describe $y=-f(x)$.
4. Find $f(3)$ if $f(x)=2x^2-1$.
5. Sketch $|x|$ then $y=|x|-2$.
6. State range of $y=(x-1)^2+5$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 10: Composite & Inverse Functions

FUNCTIONS

LEARNING GOALS

- Understand the central ideas and notation of composite & inverse functions.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Composition applies one function after another. An inverse reverses a one-to-one function. Domain restrictions are often needed before an inverse exists.

KEY FORMULAS

$$(fg)(x) = f(g(x))$$

composition

$$f^{-1}(f(x)) = x$$

inverse property

swap x, y then solve for y

inverse method

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 10: Worked Practice

FUNCTIONS

WORKED EXAMPLE

Write $y=3x-5$, swap x and y , then solve.

Find inverse of $f(x)=3x-5$

1. $x=3y-5$

2. $y=(x+5)/3$

Answer: $f^{-1}(x)=(x+5)/3$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. If $f=2x+1, g=x^2$, find fg .
2. Find gf .
3. Find inverse of $5x+2$.
4. Verify an inverse by composition.
5. Explain why x^2 on $\mathbb{R}$ has no inverse.
6. Restrict x^2 so an inverse exists.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 11: Limits

CALCULUS

LEARNING GOALS

- Understand the central ideas and notation of limits.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A limit describes the value a function approaches as x approaches a point or infinity. Limits allow calculus to handle instantaneous change even when direct substitution gives an indeterminate form.

KEY FORMULAS

$\lim_{x \rightarrow a} f(x)$

limit at a point

$0/0$

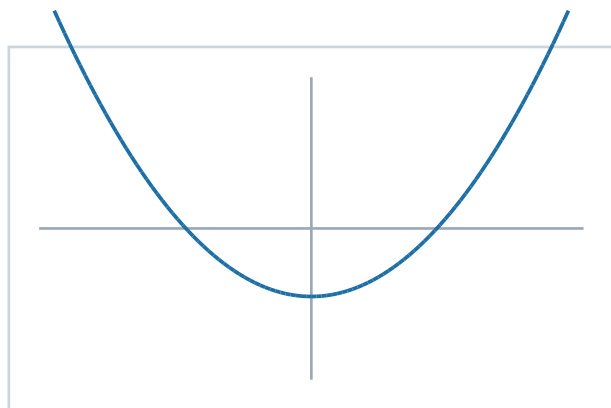
indeterminate form: simplify first

$\lim_{x \rightarrow 0} 1/x = 0$

basic infinite limit

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.



GRAPH READING

Read intercepts, turning points, asymptotes, domain/range and end behaviour directly from the graph. Connect each visible feature with its algebraic cause.

Lesson 11: Worked Practice

CALCULUS

WORKED EXAMPLE

Direct substitution gives $0/0$, so factor first.

Evaluate $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$

1. $(x^2 - 4) = (x - 2)(x + 2)$

2. Cancel $x - 2$, then substitute $x = 2$

Answer: 4

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find $\lim_{x \rightarrow 3} (x + 1)$.
2. Evaluate $\lim_{x \rightarrow 1} (x^2 - 1)/(x - 1)$.
3. Find $\lim_{x \rightarrow 5} 5/x$.
4. Evaluate $\lim_{x \rightarrow 0} \sin x/x$.
5. Identify a vertical asymptote using a limit.
6. Explain left- and right-hand limits.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 12: Continuity

CALCULUS

LEARNING GOALS

- Understand the central ideas and notation of continuity.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A function is continuous at a point when its value agrees with the limit from both sides. Continuity is crucial for modelling physical processes without sudden jumps.

KEY FORMULAS

$\lim_{x \rightarrow a} f(x) = f(a)$

continuity at $x=a$

left limit = right limit

two-sided limit exists

IVT

continuous functions take intermediate values

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 12: Worked Practice

CALCULUS

WORKED EXAMPLE

Match the left-hand limit with the function value.

Find k for continuity at $x=2$: $f=x+1$ if $x<2$, k if $x=2$

1. $\lim_{x \rightarrow 2^-} f(x) = 3$

2. Set $k=3$

Answer: $k=3$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Test continuity of $1/x$ at $x=0$.
2. Find k for a simple piecewise function.
3. Explain removable discontinuity.
4. Explain jump discontinuity.
5. State Intermediate Value Theorem.
6. Give a real-world continuous quantity.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 13: Differentiation

CALCULUS

LEARNING GOALS

- Understand the central ideas and notation of differentiation.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

The derivative measures instantaneous rate of change and the slope of a tangent. Differentiation rules let us compute derivatives efficiently without returning to the limit definition each time.

KEY FORMULAS

$\frac{d}{dx} x^n = n x^{(n-1)}$	power rule
$(uv)' = u'v + uv'$	product rule
$(\frac{u}{v})' = \frac{(u'v - uv')}{v^2}$	quotient rule

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 13: Worked Practice

CALCULUS

WORKED EXAMPLE

Differentiate each term using the power rule.

Differentiate $y=3x^4-5x^2+7$

1. $y'=12x^3-10x$

2. constant derivative is 0

Answer: $12x^3-10x$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Differentiate x^5 .
2. Differentiate $4x^3-2x$.
3. Differentiate $(x^2+1)(x-3)$.

4. Differentiate $(2x+1)/(x-1)$.
5. Differentiate $\sin x$.
6. Differentiate $e^{(3x)}$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 14: Applications of Derivatives

CALCULUS

LEARNING GOALS

- Understand the central ideas and notation of applications of derivatives.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Derivatives identify increasing/decreasing intervals, local maxima and minima, tangent lines, rates of change and optimal values. These ideas connect calculus to science and economics.

KEY FORMULAS

$f'(x)=0$ stationary point

f' changes + to - local maximum

f' changes - to + local minimum

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 14: Worked Practice

CALCULUS

WORKED EXAMPLE

Differentiate and set the derivative equal to zero.

Find turning point of $f=x^2-4x+1$

1. $f'=2x-4=0 \Rightarrow x=2$

2. $f(2)=-3$

Answer: minimum at (2,-3)

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find stationary points of x^3-3x .
2. Find tangent to $y=x^2$ at $x=2$.
3. Where is x^2-6x increasing?
4. Optimise area of a rectangle with fixed perimeter.
5. Use second derivative to classify a point.
6. Interpret velocity as derivative of position.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 15: Antiderivatives

CALCULUS

LEARNING GOALS

- Understand the central ideas and notation of antiderivatives.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Integration reverses differentiation. An indefinite integral represents a family of antiderivatives and therefore includes an arbitrary constant C.

KEY FORMULAS

$x^n \, dx = x^{(n+1)/(n+1)} + C$	$n \neq -1$
$\frac{1}{x} \, dx = \ln x + C$	logarithmic integral
$e^x \, dx = e^x + C$	exponential integral

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 15: Worked Practice

CALCULUS

WORKED EXAMPLE

Apply the power rule term by term.

Integrate $6x^2 - 4x + 3$

1. $6x^2 \, dx = 2x^3$; $-4x \, dx = -2x^2$

2. $3 \, dx = 3x$

Answer: $2x^3 - 2x^2 + 3x + C$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Integrate x^4 .
2. Integrate $3x^2 + 2$.
3. Integrate $1/x$.
4. Integrate $\cos x$.
5. Find F if $F' = 2x$ and $F(1) = 5$.
6. Check an antiderivative by differentiating.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 16: Definite Integration

CALCULUS

LEARNING GOALS

- Understand the central ideas and notation of definite integration.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A definite integral gives signed accumulation over an interval. The Fundamental Theorem of Calculus connects antiderivatives to area and total change.

KEY FORMULAS

$$\int_a^b f(x)dx = F(b)-F(a)$$

Fundamental Theorem

$$\int_a^b = -\int_b^a$$

reverse limits

$$\int_a^b f + \int_b^c f = \int_a^c f$$

additivity

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 16: Worked Practice

CALCULUS

WORKED EXAMPLE

Find an antiderivative and apply the limits.

Evaluate $\int_0^2 (3x^2+1) \, dx$

1. $F = x^3 + x$

2. $F(2) - F(0) = 8 + 2$

Answer: 10

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Evaluate $\int_0^1 x \, dx$.
2. Evaluate $\int_1^3 2x \, dx$.
3. Use symmetry for an odd function on $[-a, a]$.
4. Interpret negative signed area.
5. Find average value of f on $[a, b]$.
6. Estimate an integral using rectangles.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

CALCULUS

- Understand the central ideas and notation of applications of integration.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

Definite integrals calculate geometric area, displacement from velocity, accumulated growth and volumes. Always check whether the integrand is above or below the axis when interpreting area.

$$V = \pi [R(x)^2 - r(x)^2] dx$$
 washers

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 17: Worked Practice

CALCULUS

WORKED EXAMPLE

Integrate the non-negative function over the interval.

Area under $y=x$ from 0 to 3

1. $\text{Area} = \int_0^3 x \, dx$

2. $= \left[\frac{x^2}{2} \right]_0^3$

Answer: $\frac{9}{2}$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find area under x^2 from 0 to 2.
2. Find area between $y=x$ and $y=x^2$ on $[0,1]$.
3. Find displacement if $v=2t+1$ from 0 to 4.
4. Explain distance vs displacement.
5. Set up a volume of revolution integral.
6. Find accumulated quantity from a rate.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 18: Trigonometry & Radians

TRIGONOMETRY

LEARNING GOALS

- Understand the central ideas and notation of trigonometry & radians.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Radians measure angles using arc length and are the natural unit for calculus. Trigonometric ratios connect angles with side ratios and points on the unit circle.

KEY FORMULAS

$$\pi \text{ radians} = 180 \text{ degrees}$$

conversion

$$s = r\theta$$

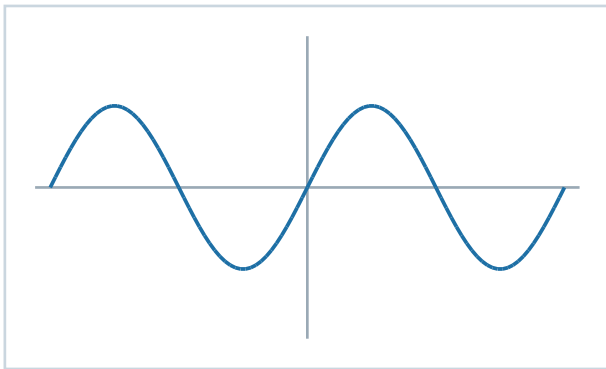
arc length

$$\sin^2 x + \cos^2 x = 1$$

Pythagorean identity

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.



GRAPH READING

Read intercepts, turning points, asymptotes, domain/range and end behaviour directly from the graph. Connect each visible feature with its algebraic cause.

Lesson 18: Worked Practice

TRIGONOMETRY

WORKED EXAMPLE

Multiply degrees by $\pi/180$.

Convert 150 degrees to radians

1. $150\pi/180$
2. simplify by 30

Answer: $5\pi/6$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Convert 60° to radians.
2. Convert $3\pi/4$ to degrees.
3. Find arc length for $r=5, \theta=1.2$.
4. Evaluate $\sin(\pi/6)$.
5. Evaluate $\cos \pi$.
6. State signs of $\sin, \cos, \tan$ by quadrant.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 19: Trigonometric Identities & Equations

TRIGONOMETRY

LEARNING GOALS

- Understand the central ideas and notation of trigonometric identities & equations.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Identities are equations true for every allowed angle. Trigonometric equations require all solutions in a specified interval, so symmetry and periodicity matter.

KEY FORMULAS

$1 + \tan^2 x = \sec^2 x$	identity
$\sin 2x = 2 \sin x \cos x$	double angle
$\cos 2x = \cos^2 x - \sin^2 x$	double angle

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 19: Worked Practice

TRIGONOMETRY

WORKED EXAMPLE

First solve $\sin x = 1/2$, then use unit-circle symmetry.

Solve $2\sin x = 1$ for $0 \leq x < 2\pi$

1. reference angle $\pi/6$
2. sine positive in quadrants I and II

Answer: $x = \pi/6, 5\pi/6$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Prove $\sin^2 x + \cos^2 x = 1$ is consistent at $x = \pi/4$.
2. Solve $\cos x = 0$ on $[0, 2\pi)$.
3. Solve $\tan x = 1$ on $[0, 2\pi)$.
4. Simplify $1 - \sin^2 x$.
5. Use double-angle formula for $\sin(2x)$.
6. Solve $2\cos^2 x - 1 = 0$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 20: Trigonometric Graphs

TRIGONOMETRY

LEARNING GOALS

- Understand the central ideas and notation of trigonometric graphs.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Sine and cosine are periodic wave functions. Parameters control amplitude, period, phase shift and vertical shift, making these models useful for oscillations, seasons and waves.

KEY FORMULAS

$$y = A \sin(Bx + C) + D$$

general sinusoid

$$\text{amplitude} = |A|$$

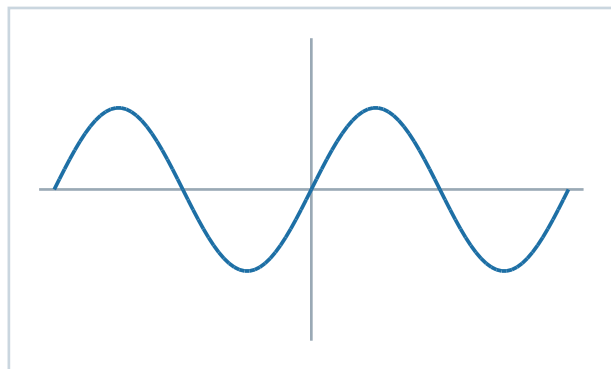
vertical scale

$$\text{period} = 2\pi / |B|$$

cycle length

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.



GRAPH READING

Read intercepts, turning points, asymptotes, domain/range and end behaviour directly from the graph. Connect each visible feature with its algebraic cause.

Lesson 20: Worked Practice

TRIGONOMETRY

WORKED EXAMPLE

Read each parameter directly.

Analyse $y=3\sin(2x)-1$

1. amplitude 3; period π
2. vertical shift -1

Answer: range $[-4,2]$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. State amplitude of $5\cos x$.
2. Find period of $\sin(4x)$.
3. Find range of $2\sin x+3$.
4. Sketch one period of $\sin x$.
5. Compare $\sin x$ and $\cos x$.
6. Model a periodic quantity with a sinusoid.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 21: Coordinate Geometry

GEOMETRY

LEARNING GOALS

- Understand the central ideas and notation of coordinate geometry.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Coordinate geometry turns geometric relationships into algebra. Slope, distance, midpoint and circle equations support analytic proofs and modelling.

KEY FORMULAS

$$m = (y_2 - y_1) / (x_2 - x_1)$$

slope

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

distance

$$(x - a)^2 + (y - b)^2 = r^2$$

circle

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 21: Worked Practice

GEOMETRY

WORKED EXAMPLE

Use point-slope form.

Equation of line through (1,2) with slope 3

1. $y-2=3(x-1)$

2. expand: $y=3x-1$

Answer: $y=3x-1$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find slope through (2,5),(6,13).
2. Find midpoint of (-2,4),(8,0).
3. Find distance between (0,0),(3,4).
4. Equation of line with slope -2 through (3,1).
5. Find centre and radius of $x^2+y^2-4x+6y=3$.
6. Test whether two lines are perpendicular.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 22: Vectors in 2D & 3D

VECTORS

LEARNING GOALS

- Understand the central ideas and notation of vectors in 2d & 3d.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Vectors describe quantities with magnitude and direction. Components make vector addition, scalar multiplication, lengths and dot products straightforward.

KEY FORMULAS

$ v = \sqrt{v_1^2 + v_2^2 + v_3^2}$	magnitude
$u \cdot v = u v \cos \theta$	dot product
$u \cdot v = 0$	perpendicular vectors

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 22: Worked Practice

VECTORS

WORKED EXAMPLE

Use the dot-product formula.

Find angle between $u=(1,0)$ and $v=(1,1)$

1. $u \cdot v = 1$; $|u| = 1$; $|v| = \sqrt{2}$

2. $\cos \theta = 1/\sqrt{2}$

Answer: $\theta = 45^\circ$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find $|(3,4)|$.
2. Add $(2,-1)+(5,3)$.
3. Find $2(1,-2,4)$.
4. Compute $(1,2) \cdot (3,-1)$.
5. Test perpendicularity of $(2,1), (-1,2)$.
6. Find unit vector in direction $(3,4)$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 23: Lines & Planes

VECTORS

LEARNING GOALS

- Understand the central ideas and notation of lines & planes.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Vector equations describe lines and planes in space. Parametric forms show how points move along a direction; normal vectors describe plane orientation.

KEY FORMULAS

$r=a+t\ d$	vector line
$n\cdot(r-a)=0$	plane through a with normal n
$n_1\ parallel\ n_2$	parallel planes

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 23: Worked Practice

VECTORS

WORKED EXAMPLE

Insert point and direction into vector form.

Line through $A=(1,2,0)$ in direction $d=(2,-1,3)$

1. $r=(1,2,0)+t(2,-1,3)$

2. component equations follow directly

Answer: $x=1+2t, y=2-t, z=3t$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Write line through origin direction $(1,2,3)$.
2. Check if point lies on a parametric line.
3. Write plane normal $(1,0,2)$ through $(3,1,-1)$.
4. Test parallel planes.
5. Find angle between direction vectors.
6. Describe intersection possibilities of two planes.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 24: Complex Numbers

COMPLEX NUMBERS

LEARNING GOALS

- Understand the central ideas and notation of complex numbers.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Complex numbers extend the real number system by introducing i with $i^2=-1$. They allow every polynomial equation to have roots in the complex number system.

KEY FORMULAS

$z=a+bi$	rectangular form
$ z =\sqrt{a^2+b^2}$	modulus
$z^* = a-bi$	conjugate

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 24: Worked Practice

COMPLEX NUMBERS

WORKED EXAMPLE

Move 4 to the other side and use $i^2 = -1$.

Solve $x^2 + 4 = 0$

1. $x^2 = -4$

2. $x = \pm 2i$

Answer: $x = \pm 2i$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Simplify $i^2 + i^4$.
2. Add $(2 + 3i) + (1 - 5i)$.
3. Multiply $(1 + i)(2 - i)$.
4. Find modulus of $3 - 4i$.
5. Find conjugate of $-2 + 7i$.
6. Solve $x^2 - 2x + 5 = 0$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 25: Sequences

DISCRETE MATHEMATICS

LEARNING GOALS

- Understand the central ideas and notation of sequences.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A sequence is an ordered list generated by a rule. Arithmetic sequences have constant difference; geometric sequences have constant ratio. Both appear in growth and finance models.

KEY FORMULAS

$a_n=a_1+(n-1)d$	arithmetic term
$a_n=a_1 r^{(n-1)}$	geometric term
$r=a_{(n+1)}/a_n$	common ratio

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 25: Worked Practice

DISCRETE MATHEMATICS

WORKED EXAMPLE

This is arithmetic with $a_1=5, d=3$.

Find 20th term of 5,8,11,...

1. $a_{20}=5+19(3)$

2. $=62$

Answer: 62

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

- Find 10th term of 2,5,8,...
- Find common ratio of 3,6,12,...
- Write formula for 7,14,21,...
- Find n if $4+3(n-1)=40$.
- Determine whether 2,6,18 is arithmetic or geometric.
- Model 5% annual growth geometrically.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 26: Series & Binomial Theorem

DISCRETE MATHEMATICS

LEARNING GOALS

- Understand the central ideas and notation of series & binomial theorem.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A series is the sum of sequence terms. Closed formulas save time when many terms are involved. The binomial theorem expands powers such as $(a+b)^n$ using combinations.

KEY FORMULAS

$S_n = \frac{n}{2}[2a + (n-1)d]$	arithmetic sum
$S_n = \frac{a(1-r^n)}{(1-r)}$	finite geometric sum
$(a+b)^n = \sum C(n,k) a^{n-k} b^k$	binomial theorem

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 26: Worked Practice

DISCRETE MATHEMATICS

WORKED EXAMPLE

Use arithmetic series with $a=3, d=4$.

Sum first 20 terms of 3, 7, 11, ...

1. $S_{20} = 20/2[6 + 19(4)]$

2. $= 10(82)$

Answer: 820

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Sum $1+2+\dots+100$.
2. Find sum first 6 terms of 3, 6, 12, ...
3. Expand $(x+1)^4$.
4. Find coefficient of x^2 in $(x+2)^4$.
5. Find infinite geometric sum $6+3+1.5+\dots$
6. State condition for an infinite geometric series to converge.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 27: Combinatorics & Probability

PROBABILITY

LEARNING GOALS

- Understand the central ideas and notation of combinatorics & probability.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Counting methods prevent missed or double-counted outcomes. Probability then measures the relative likelihood of events. Conditional probability updates likelihood using known information.

KEY FORMULAS

$n!=n(n-1)...1$	factorial
$C(n,r)=n!/[r!(n-r)!]$	combinations
$P(A B)=P(AB)/P(B)$	conditional probability

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 27: Worked Practice

PROBABILITY

WORKED EXAMPLE

Order does not matter, so use combinations.

Choose 3 students from 8

1. $C(8,3) = 8!/(3!5!)$

2. $=56$

Answer: 56

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Compute $6!$.
2. Find $C(10,2)$.
3. Probability of head on fair coin.
4. Probability sum 7 with two dice.
5. If $P(A)=0.4, P(B)=0.5$ independent, find $P(AB)$.
6. Use complement to find probability of at least one success.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 28: Random Variables & Distributions

PROBABILITY

LEARNING GOALS

- Understand the central ideas and notation of random variables & distributions.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

A random variable assigns numerical values to outcomes. Probability distributions describe the possible values and their probabilities; expectation and variance summarise centre and spread.

KEY FORMULAS

$E(X)=\sum xP(X=x)$

expected value

$Var(X)=E(X^2)-[E(X)]^2$

variance

$P(X=k)=C(n,k)p^k(1-p)^{(n-k)}$

binomial model

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 28: Worked Practice

PROBABILITY

WORKED EXAMPLE

Multiply each value by its probability and add.

For $X=0,1,2$ with probabilities $.2, .5, .3$, find $E(X)$

1. $0(.2)+1(.5)+2(.3)$

2. $.5+.6$

Answer: 1.1

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Check probabilities $.2, .3, .5$ form a distribution.
2. Find $E(X)$ for a small table.
3. Find binomial $P(X=2)$ with $n=4, p=.5$.
4. State mean of $\text{Bin}(n, p)$.
5. State conditions for binomial model.
6. Interpret standard deviation.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 29: Statistics & Regression

STATISTICS

LEARNING GOALS

- Understand the central ideas and notation of statistics & regression.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Statistics summarises data and studies relationships. Correlation measures the strength of linear association, while regression provides an equation for prediction within a reasonable data range.

KEY FORMULAS

$$\text{mean} = \Sigma x / n$$

arithmetic mean

$$\text{ssqrt}(\Sigma(x - \bar{x})^2 / (n - 1))$$

sample standard deviation

$$y = a + bx$$

linear regression

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 29: Worked Practice

STATISTICS

WORKED EXAMPLE

Add values and divide by 4.

Find mean of 4,7,8,11

1. $\text{sum}=30$

2. $30/4=7.5$

Answer: 7.5

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Find median of 2,5,8,9,12.
2. Find mean of 3,3,6,8.
3. Find range of 4,10,7,2.
4. Interpret $r=0.92$.
5. Explain why correlation is not causation.
6. Predict y from $y=2.5x+4$ at $x=6$.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Lesson 30: Numerical Methods & Mathematical Modelling

MODELLING

LEARNING GOALS

- Understand the central ideas and notation of numerical methods & mathematical modelling.
- Apply standard formulas or methods accurately.
- Explain restrictions, assumptions and checks that make a solution valid.

CORE IDEA

Many equations have no simple exact solution. Numerical methods approximate roots and values. Mathematical modelling turns a real situation into variables, assumptions, equations and testable predictions.

KEY FORMULAS

Newton: $x_{(n+1)} = x_n - f(x_n)/f'(x_n)$

root iteration

error=approx-exact

absolute error idea

model cycle

define -> solve -> validate -> refine

METHOD CHECKLIST

- Identify the mathematical structure before choosing a rule.
- Write each algebraic step clearly; do not skip sign changes or restrictions.
- Substitute the result back, differentiate, or estimate numerically when a check is available.

Study focus	What to do
Definition	Know exactly what the term means.
Condition	Check when the rule is allowed.
Representation	Move between equation, table and graph.
Verification	Use an independent check where possible.

Lesson 30: Worked Practice

MODELLING

WORKED EXAMPLE

Use $f' = 2x$.

One Newton step for $f = x^2 - 2$ from $x_0 = 1.5$

1. $x_1 = 1.5 - (2.25 - 2)/3$

2. $\approx 1.416666\dots$

Answer: $x \approx 1.4167$

COMMON ERROR

Do not apply a memorised rule mechanically. Check signs, brackets, domains, units and whether every solution is permitted by the original problem.

PRACTICE SET

1. Use midpoint of $[1, 2]$ to start bisection for $x^2 - 2$.
2. Perform one Newton step for $x^2 - 5$ from $x_0 = 2$.
3. Explain rounding error.
4. State one modelling assumption for population growth.
5. Explain validation using data.
6. Compare exact and numerical solutions.

SELF CHECK

Can you explain why your method works? Can you estimate the answer before calculating? Can you identify a second method or a graphical interpretation?

Mixed Review 1 - Algebra & Functions

REVISION

INSTRUCTIONS

Work without notes first. Show every important algebraic step. Use a calculator only when necessary, and give exact answers where appropriate.

1. Factorise $x^2 - 5x + 6$.

2. Solve $2x^2 + x - 6 = 0$.

3. Find inverse of $f(x) = 4x - 7$.

4. State domain of $(x+1)/(x-3)$.

5. Solve $|2x - 1| < 5$.

6. Find $P(2)$ for $P = x^3 - x + 1$.

7. A model has $y = 3(1.08)^t$. Identify initial value and growth rate.

8. Sketch $y = (x-1)^2 - 4$ and label vertex.

9. Solve $\ln x = 2$.

10. Find determinant of $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$.

Mixed Review 2 - Calculus

REVISION

INSTRUCTIONS

Work without notes first. Show every important algebraic step. Use a calculator only when necessary, and give exact answers where appropriate.

1. Evaluate $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$.

2. Differentiate $5x^4 - 3x^2 + 2$.

3. Find stationary points of $x^3 - 3x$.

4. Integrate $4x^3 - 2x$.

5. Evaluate $\int_0^2 (x+1) dx$.

6. Find tangent to $y = x^2 + 1$ at $x = 1$.

7. A particle has $s = t^3 - 6t^2 + 9t$. Find velocity.

8. Use derivative to minimise $x^2 - 8x + 20$.

9. Find area under $y = 2x$ from 1 to 4.

10. Explain continuity at $x = a$.

Mixed Review 3 - Trigonometry & Geometry

REVISION

INSTRUCTIONS

Work without notes first. Show every important algebraic step. Use a calculator only when necessary, and give exact answers where appropriate.

1. Convert 225° to radians.

2. Solve $\sin x = 3/2$ on $[0, 2\pi)$.

3. Find amplitude and period of $4\cos(3x)$.

4. Find distance between $(2, -1)$ and $(5, 3)$.

5. Find equation of line slope 2 through $(1, 4)$.

6. Compute dot product $(2, 1) \cdot (3, -4)$.

7. Find angle between perpendicular vectors.

8. Write circle centre $(2, -3)$, radius 5.

9. Use $\sin^2 x + \cos^2 x = 1$ to find $\cos x$ if $\sin x = 3/5$ and x acute.

10. Write vector line through $(0, 1, 2)$ direction $(2, -1, 1)$.

Mixed Review 4 - Complex, Sequences & Probability

REVISION

INSTRUCTIONS

Work without notes first. Show every important algebraic step. Use a calculator only when necessary, and give exact answers where appropriate.

1. Simplify $(2+3i)+(4-i)$.

2. Find $|5-12i|$.

3. Find 15th term of arithmetic sequence 4,7,10,...

4. Find sum first 10 terms of 2,5,8,...

5. Compute $C(9,3)$.

6. Find $P(\text{at least one head})$ in two fair tosses.

7. For $\text{Bin}(5,0.4)$, write expression for $P(X=2)$.

8. Find expected value for X values 0,1 with $P=.3,.7$.

9. Expand $(1+x)^5$ up to x^3 .

10. Find infinite sum $8+4+2+\dots$

Mixed Review 5 - Statistics & Modelling

REVISION

INSTRUCTIONS

Work without notes first. Show every important algebraic step. Use a calculator only when necessary, and give exact answers where appropriate.

1. Find mean of 5,7,9,9,10.

2. Find median of 3,4,8,12,15,20.

3. Interpret correlation $r=-0.85$.

4. Predict y if $y=1.8x+2.4$ and $x=5$.

5. Perform one Newton step for x^2-3 from $x_0=2$.

6. Explain why correlation does not prove causation.

7. Describe one source of sampling bias.

8. State one assumption in exponential growth modelling.

9. Explain what validation means in a model.

10. Compare exact and approximate answers.

Mock Examination A - Pure Mathematics

ASSESSMENT

Suggested time: 75 minutes. Show reasoning. Exact answers are preferred unless a decimal is requested.

Question 1

Factorise and solve $x^3 - 4x = 0$.

Question 2

Find inverse of $f(x) = (2x-3)/5$.

Question 3

Solve $2^x = 13$ to three decimal places.

Question 4

Analyse domain and asymptotes of $(2x+1)/(x-3)$.

Question 5

Solve $|3x-2| \leq 7$.

Question 6

Find roots of $x^2 - 6x + 13$ in C.

Question 7

Find 12th term and first 12-term sum of 5, 9, 13, ...

Question 8

Expand $(1-2x)^4$.

Mock Examination B - Calculus & Geometry

ASSESSMENT

Suggested time: 75 minutes. Show reasoning. Exact answers are preferred unless a decimal is requested.

Question 1

Evaluate a rational limit producing $0/0$.

Question 2

Differentiate $y=(x^2+1)e^x$.

Question 3

Find and classify stationary points of x^3-6x^2+9x .

Question 4

Evaluate a definite integral of a polynomial.

Question 5

Find area between $y=x$ and $y=x^2$ on $[0,1]$.

Question 6

Solve a trigonometric equation on $[0,2\pi)$.

Question 7

Find equation of a line and circle from geometric data.

Question 8

Use vectors to test perpendicularity.

Mock Examination C - Probability & Statistics

ASSESSMENT

Suggested time: 75 minutes. Show reasoning. Exact answers are preferred unless a decimal is requested.

Question 1

Count 4-person committees from 11 people.

Question 2

Compute a conditional probability from a two-event model.

Question 3

For $X \sim \text{Bin}(8, 0.3)$, find $P(X=2)$.

Question 4

Find $E(X)$ from a discrete probability table.

Question 5

Find mean, median and range of a sample.

Question 6

Interpret $r = -0.72$ in context.

Question 7

Use a regression equation for interpolation and explain extrapolation risk.

Question 8

Explain one cause of bias in a survey.

Mock Examination D - Comprehensive Scientific Mathematics

ASSESSMENT

Suggested time: 75 minutes. Show reasoning. Exact answers are preferred unless a decimal is requested.

Question 1

Solve a quadratic and interpret the discriminant.

Question 2

Find a function composition and inverse.

Question 3

Evaluate a limit and test continuity of a piecewise function.

Question 4

Differentiate, optimise and interpret a physical quantity.

Question 5

Integrate a rate to obtain total change.

Question 6

Model a periodic phenomenon with a sinusoid.

Question 7

Use arithmetic/geometric sequences in a finance context.

Question 8

Perform one Newton iteration and discuss numerical error.

Quick Answers to Practice Questions 1-3

ANSWERS

These are short answers for checking. Full marks in an examination require method and reasoning, not answers alone.

Lesson 01: Algebraic Expressions & Factorisation

1) $2x^2+7x-15$ 2) $(x-3)(x+3)$ 3) $(2x+3)^2$

Lesson 02: Linear & Quadratic Equations

1) $x=4$ 2) $x=3,4$ 3) $x=1/3,-1$

Lesson 03: Systems of Equations & Matrices

1) $x=3,y=3$ 2) 10 3) $\begin{bmatrix} 13 \\ 5 \end{bmatrix}$

Lesson 04: Inequalities & Absolute Value

1) $x>4$ 2) x^2 3) $1<x<7$

Lesson 05: Polynomial Functions

1) 3 2) No: $P(2)=4$ 3) $(x-3)(x-2)(x+2)$

Lesson 06: Rational Functions

1) $x \pm 3$ 2) $x=-4$ 3) $x=5$

Lesson 07: Exponents & Radicals

1) x^8 2) x^{-4} 3) 8

Lesson 08: Logarithms

1) $2^3=8$ 2) 3 3) $\ln 3+2\ln x$

Lesson 09: Functions & Transformations

1) x^1 2) left 4 3) reflection in x-axis

Lesson 10: Composite & Inverse Functions

1) $2x^2+1$ 2) $(2x+1)^2$ 3) $(x-2)/5$

Quick Answers to Practice Questions 1-3

ANSWERS

These are short answers for checking. Full marks in an examination require method and reasoning, not answers alone.

Lesson 11: Limits

1) 4 2) 2 3) 0

Lesson 12: Continuity

1) not continuous 2) match one-sided limits 3) hole

Lesson 13: Differentiation

1) $5x^4$ 2) $12x^2-2$ 3) $2x(x-3)+(x^2+1)$

Lesson 14: Applications of Derivatives

1) $x=\pm 1$ 2) $y=4x-4$ 3) $x>3$

Lesson 15: Antiderivatives

1) $x^5/5+C$ 2) x^3+2x+C 3) $\ln|x|+C$

Lesson 16: Definite Integration

1) $1/2$ 2) 8 3) 0

Lesson 17: Applications of Integration

1) $8/3$ 2) $1/6$ 3) 20

Lesson 18: Trigonometry & Radians

1) $\pi/3$ 2) 135° 3) 6

Lesson 19: Trigonometric Identities & Equations

1) 1 2) $\pi/2, 3\pi/2$ 3) $\pi/4, 5\pi/4$

Lesson 20: Trigonometric Graphs

1) 5 2) $\pi/2$ 3) $[1,5]$

Quick Answers to Practice Questions 1-3

ANSWERS

These are short answers for checking. Full marks in an examination require method and reasoning, not answers alone.

Lesson 21: Coordinate Geometry

1) 2 2) (3,2) 3) 5

Lesson 22: Vectors in 2D & 3D

1) 5 2) (7,2) 3) (2,-4,8)

Lesson 23: Lines & Planes

1) $r=t(1,2,3)$ 2) substitute parameters 3) $(r-(3,1,-1)) \cdot (1,0,2)=0$

Lesson 24: Complex Numbers

1) 0 2) $3-2i$ 3) $3+i$

Lesson 25: Sequences

1) 29 2) 2 3) $a_n=7n$

Lesson 26: Series & Binomial Theorem

1) 5050 2) 189 3) $x^4+4x^3+6x^2+4x+1$

Lesson 27: Combinatorics & Probability

1) 720 2) 45 3) $1/2$

Lesson 28: Random Variables & Distributions

1) yes 2) use $\sum xP(x)$ 3) $6/16=3/8$

Lesson 29: Statistics & Regression

1) 8 2) 5 3) 8

Lesson 30: Numerical Methods & Mathematical Modelling

1) 1.5 2) 2.25 3) depends on representation

Essential Formula Sheet

REFERENCE

ALGEBRA

- Quadratic: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Arithmetic: $a_n = a_1 + (n-1)d$; $S_n = \frac{n}{2}[2a + (n-1)d]$
- Geometric: $a_n = a_1 r^{(n-1)}$; $S_n = \frac{a(1-r^n)}{(1-r)}$

CALCULUS

- $\frac{d}{dx} x^n = nx^{(n-1)}$
- $(uv)' = u'v + uv'$
- $\int x^n dx = \frac{x^{(n+1)}}{(n+1)} + C$
- $\int_a^b f = F(b) - F(a)$

TRIGONOMETRY

- $\sin^2 x + \cos^2 x = 1$
- $\sin 2x = 2 \sin x \cos x$
- $\cos 2x = \cos^2 x - \sin^2 x$
- $s = r\theta$

GEOMETRY & VECTORS

- $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$
- $u \cdot v = |u| |v| \cos \theta$
- $(x-a)^2 + (y-b)^2 = r^2$

PROBABILITY & STATISTICS

- $C(n, r) = \frac{n!}{r!(n-r)!}$
- $P(A | B) = \frac{P(AB)}{P(B)}$
- $E(X) = \sum xP(X=x)$
- $\text{Var}(X) = E(X^2) - [E(X)]^2$

REMEMBER

A formula is useful only when its assumptions are satisfied. Check domain restrictions, units, calculator angle mode, and whether exact or approximate form is expected.

Glossary & Examination Strategy

REFERENCE

Term	Meaning
asymptote	line approached by a curve
derivative	instantaneous rate of change
integral	accumulated change / signed area
domain	allowed input values
range	possible output values
vector	quantity with magnitude and direction
expected value	long-run mean of a random variable
correlation	strength/direction of linear association
iteration	repeated numerical process
model	mathematical representation of a situation

EXAMINATION STRATEGY

- Read the command word: solve, prove, show, estimate, interpret and hence require different responses.
- Write exact values first; convert to decimals only when asked or useful.
- For graphs, label axes, key points, intercepts and asymptotes.
- For calculus, state the derivative/integral before substituting values.
- For probability, define events and check probabilities sum to 1.
- If stuck, write a correct formula or intermediate relationship: method marks often matter.

Scientific Mathematics Mastery Checklist

FINAL REVIEW

- ☐ I can manipulate and factorise algebraic expressions.
- ☐ I can solve equations, inequalities and systems.
- ☐ I can analyse polynomial, rational, exponential and logarithmic functions.
- ☐ I can work with limits and continuity.
- ☐ I can differentiate and solve optimisation problems.
- ☐ I can integrate and interpret accumulated change.
- ☐ I can solve trigonometric equations and interpret sinusoidal graphs.
- ☐ I can use coordinate geometry and vectors in 2D/3D.
- ☐ I can calculate with complex numbers.
- ☐ I can model arithmetic and geometric sequences and series.
- ☐ I can use combinatorics, conditional probability and binomial models.
- ☐ I can summarise data, interpret correlation and use regression carefully.
- ☐ I can apply a numerical method and discuss approximation error.
- ☐ I can explain my reasoning clearly, not only produce a final answer.

FINAL STUDY ADVICE

Mathematics improves through active practice. Rework problems you got wrong, explain solutions aloud, and mix topics so you learn to choose methods rather than merely recognise chapter patterns.